

BEYOND NODAL DAMPING

LOCAL TUNING. COLLECTIVE EFFECTS.

Exact interaction law, certified counterexample and repair for delay-aware SG-to-GFL co-design
Jorge Luis Mayorga Taborda | Universidad de los Andes, Colombia

1 INTRO AND OBJECTIVES

Introduction

Inverters (GFL) are replacing synchronous generators (SG). Each one locks to the grid with a delayed phase-locked loop (PLL), tuned through its gains K_p, K_i until its own oscillation decays fast enough.

Objectives

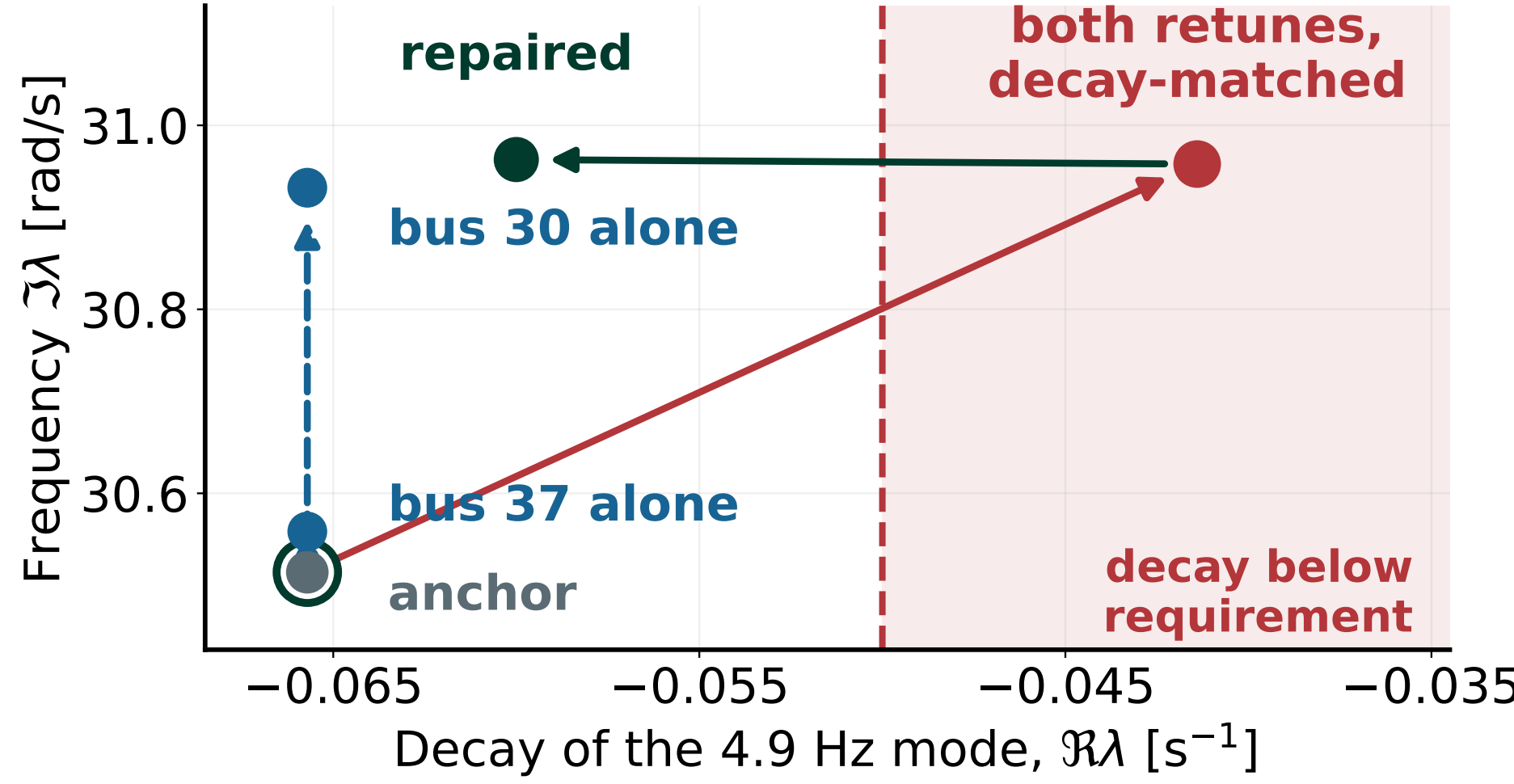
- 1 Decide whether site-by-site PLL retunes stay safe together.
- 2 Find the exact law behind any failure.
- 3 Give a repair rule and check it on the delayed nonlinear grid.

IN WORDS If every retune is safe on its own, is the grid still safe when two of them happen together?

Setting. IEEE-39, PLL delay 40 ms at every site, five frozen nonlinear events. Replaced share: $P_{\text{GFL}} = \sum_i P_i^0 \rho_i$, with ρ_i the fraction of site i replaced.

2 WHAT WE FOUND: TWO SAFE RETUNES, ONE UNSAFE GRID

Bus 30 and 37 each replace a little more SG ($\rho : 0.875 \rightarrow 0.885$, +7.9 MW GFL in total). At each site we adjust the gains so that the shared 4.9 Hz mode keeps its decay rate, 0.0657 s^{-1} .



0.0243

s^{-1} of decay lost to the interaction (certified $\pm 4 \times 10^{-7}$)

87.6%

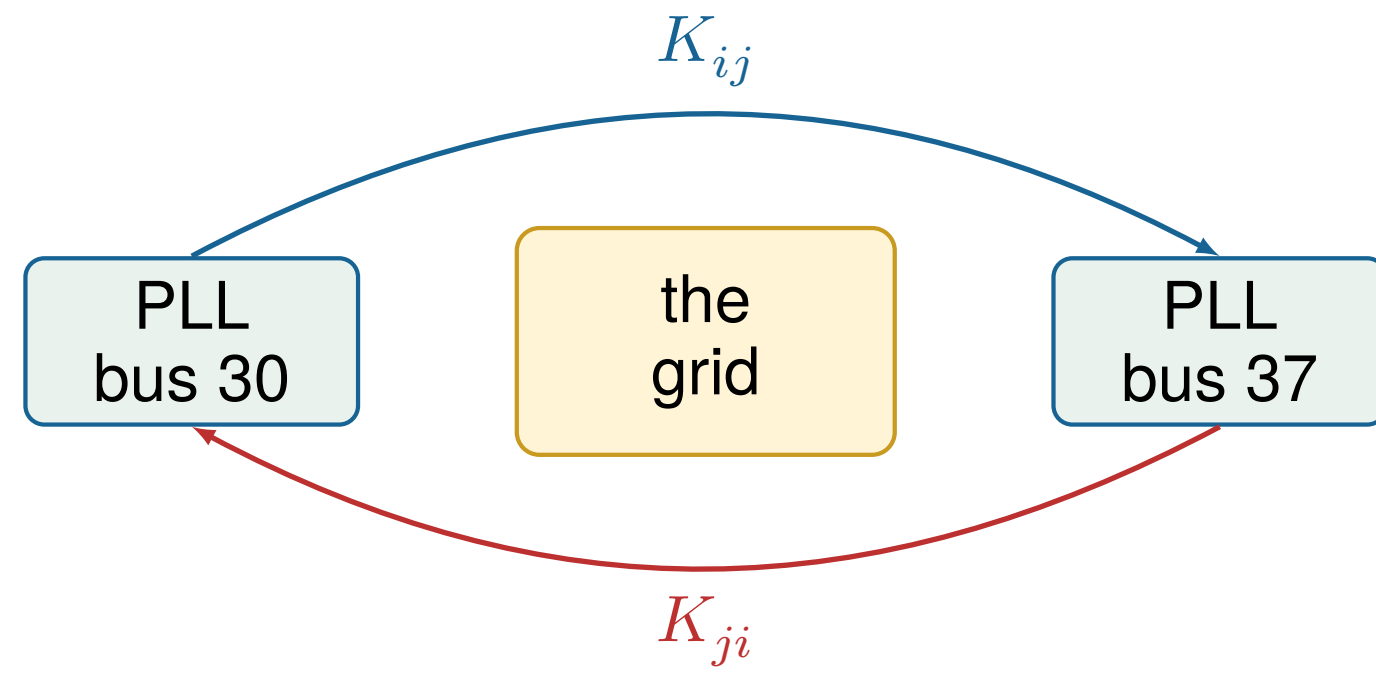
GFL share after the repair:
4,735 MW inverter, 667 MW SG kept

$$\mathcal{J} = \Re \lambda_{\text{both}} - \Re \lambda_{30} - \Re \lambda_{37} + \Re \lambda_{\text{anchor}} = +0.0243 \text{ s}^{-1}$$

IN WORDS Each site fixes the decay as if the other were not changing, and the two fixes do not add up: together the decay falls to 0.0414, below the 0.05 required. The mode is still stable; only its margin is lost. Matching the whole pole instead (decay and frequency) at each site keeps it at 0.0657.

3 WHY: THE GRID LOOP

A retune at one site changes what the other site feels through the grid. One lap around the loop has gain $P = K_{ij}K_{ji}$.



$$\frac{\det F \det F_0}{\det F_i \det F_j} = \det(I - P)$$

IN WORDS The joint determinant is the product of the single-site ones times $\det(I - P)$. If $P = 0$ the retunes are independent; otherwise the loop adds an exact interaction.

1 RETUNE
decay held at each site



2 COUPLE
 $P = K_{ij}K_{ji}$



3 INTERACT
 $\mathcal{J} = z - z_0 - \sum_i (z_i - z_0)$



4 REPAIR
 $t_{k+1} = \max\{0, \mathcal{J}(t_k) - b\}$

4 HOW: A FORMULA FOR THE GAINS AND A REPAIR RULE

A. Gains from a formula, not a search

Choose the pole λ and a pattern q . The PI term must satisfy $(K_p \lambda + K_i) y_i = \lambda^2 (1 + t_f \lambda) e^{\lambda \tau_i} q_i$. With W_i the right side over y_i :

$$K_{p,i} = \frac{\Im W_i}{\Im \lambda}$$
$$K_{i,i} = \Re W_i - \Re \lambda K_{p,i}$$

$y = G(\lambda, p)q$: how the grid returns each site's phase error. Needs invertible hidden blocks and gains inside the bounds.

IN WORDS Choose the pole you want and the gains follow. If two retunes interact, add decay until the interaction no longer eats the margin.

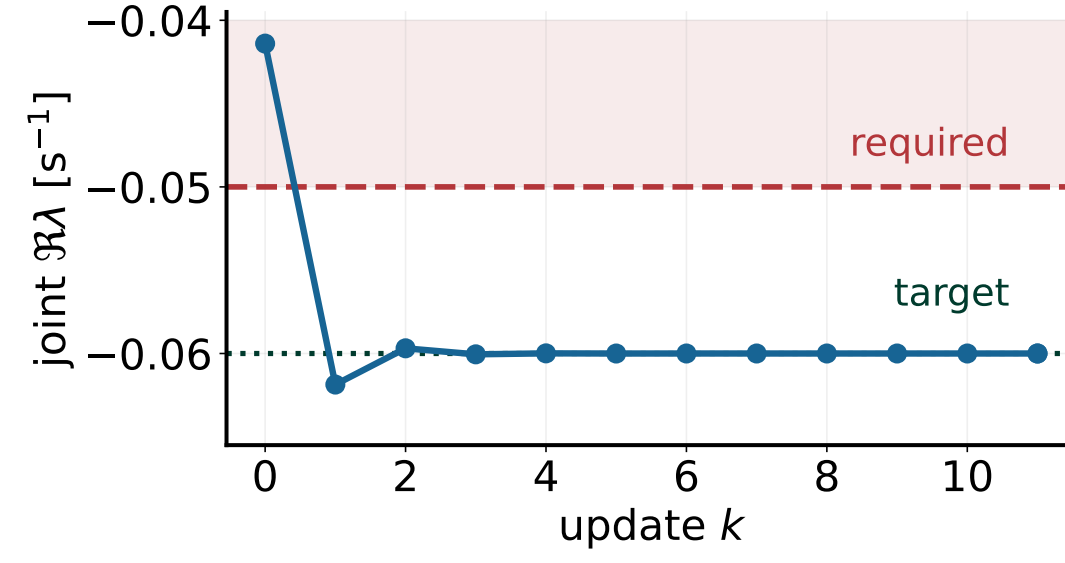
B. Repair rule when retunes interact

- 1 Retune each site alone so the mode keeps its decay.
- 2 Evaluate the joint pole on the full delayed model: this gives $\mathcal{J}$.
- 3 Give each site $t/2$ extra decay and repeat:

$$t_{k+1} = \max\{0, \mathcal{J}(t_k) - b\}, \quad b = -\alpha_{\text{anchor}} - \sigma$$

If $\sup |\mathcal{J}'| \leq L < 1$ the update contracts: $|t_k - t^*| \leq L^k |t_0 - t^*|$.

Result of the rule



11 updates: $-0.0414 \rightarrow -0.0600$. Only K_i moves (272.42 and 288.70); K_p fixed. Sampled slope ≤ 0.385 .

5 RESULTS: THE REPAIR HOLDS ON THE NONLINEAR GRID

Repaired design, full retained-device DAE with true delay (method of steps, Julia Rodas5P), 60 s, frozen impedance-load events.

Event	$ \Delta f _{\max}$ [Hz]	RoCoF [Hz/s]	V [pu]	SG slack
bus 8, −100 MW	0.457	0.145	0.988–1.078	0.0993
bus 16, +100 MW	0.439	0.129	0.972–1.061	0.0089
bus 16, −100 MW	0.466	0.131	0.986–1.079	0.0962
bus 29, +100 MW	0.423	0.138	0.973–1.061	0.0137
bus 29, −100 MW	0.436	0.219	0.983–1.075	0.1055

5/5

events pass

limits: 0.5 Hz, 0.5 Hz/s, V in [0.9, 1.1], slack ≥ 0.002

$\Re \lambda : -0.0657 \xrightarrow{\text{both retunes}} -0.0414 \xrightarrow{\text{repair}} -0.0600$

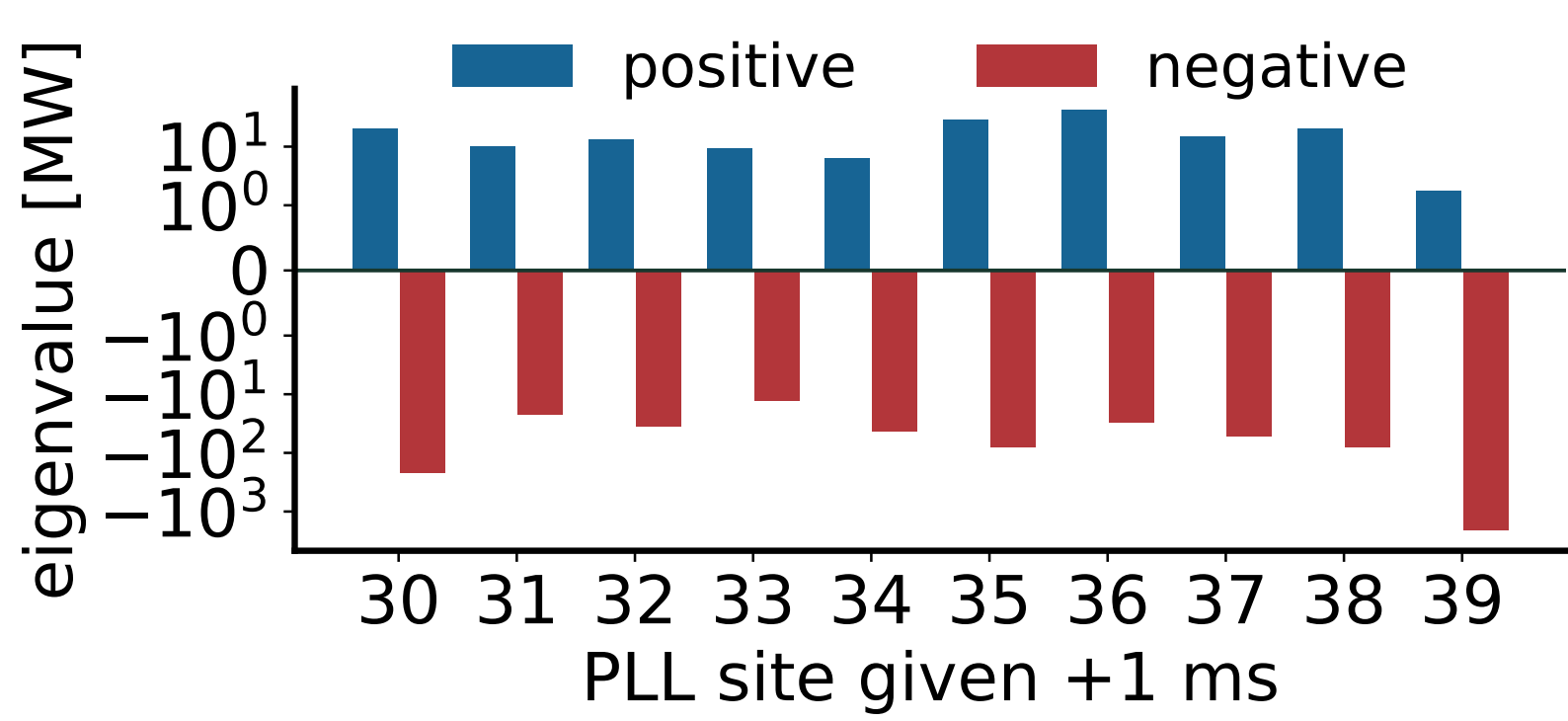
The full-spectrum margin check is a floating-point contour (0 roots beyond -0.05), not an interval certificate. No converter current-limit claim.

7 BEYOND NODAL: ONE DELAY MOVES DAMPING AROUND

Delay one PLL by 1 ms. The damping the grid supplies at frequency Ω does not just rise or fall:

$$\delta Z = \gamma u v^*$$
$$\delta D_H = \frac{1}{2} (a v^* + v a^*)$$

IN WORDS Along one network path damping grows, along another it shrinks: always one positive and one negative eigenvalue (60 of 60 checks).



IEEE-39, 5 Hz, +1 ms at each PLL site; error $< 5 \times 10^{-8}$.

Z : torque–speed impedance seen by the generators. $D_H = \text{Herm } Z(j\Omega)$: damping supplied at Ω . u, v : the two network paths from the delayed PLL; $a = \gamma u$.

Why “beyond nodal”: in one frozen design the nodal term of the projected damping is +38.7 but the collective term is −737.5; ignoring the grid paths flips the sign. **Caution:** damping at one frequency is not a stability test.

9 CONCLUSIONS: WHAT IS ESTABLISHED, WHAT IS OPEN

Exact

pair identity, rank-two law, gain formula

Certified

the pair's roots, boxes of radius 10^{-7}

Validated

5/5 delayed nonlinear events

Still open

maximum replacement; how often this happens

Retracted: the 94.2% GFL candidate fails bus-8 RoCoF (0.513 Hz/s) and bus-29 frequency (0.589 Hz). **Open:** one certified pair; decay-only retunes violated the margin in 2 of 88 resolved cases. **Not claimed:** a maximum, superiority over fixed gains (they also pass the events), instability (the pole stays stable), full-spectrum interval certificates, converter current limits.

TAKEAWAY

SAFE ALONE CAN FAIL TOGETHER.

Decay-matched retunes interact through the grid. The exact interaction sets the margin that repairs the design.

REFERENCES

- [1] Marković et al., IEEE Trans. Power Syst., 2021.
- [2] Bindel & Hood, SIAM J. Matrix Anal. Appl., 2013.
- [3] Beyn, Linear Algebra Appl., 2012.
- [4] Johansson, IEEE Trans. Comput., 2017.
- [5] Coletta & Jacquod, Phys. Rev. E, 2016.
- [6] Poolla et al., IEEE Trans. Autom. Control, 2017.

Topics: [1] small-signal stability; [2], [3] nonlinear eigenvalues; [4] interval arithmetic; [5] network connectivity and stability; [6] inertia placement.

CONTACT

Jorge Luis Mayorga Taborda

j.l.mayorga@uniandes.edu.co
Universidad de los Andes
Bogotá, Colombia

IEEE IAS Annual Meeting 2026
Vancouver, Canada